

Roshan Ganesh

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EDUCATION

University of Washington - Seattle

Graduating 2028

- *BS in* Electrical and Computer Engineering | *Cumulative GPA:* 4.0/4.0
- *Relevant Coursework:* EE 271 (Digital Circuits and Systems), EE 215 (Intro to EE) Calculus/Math 124-6, Physics 141-3 CSE 121-3 (OOP, Data Structures)
- Currently working as a Teaching Assistant (TA) for CSE 123 (Data Structures & Algorithms)

EXPERIENCE

Software Developer, [Seema's Cafe](#), Sammamish, WA | [Github](#)

Sept. 2025 - Mar. 2026

- Designed and developed a website for a local cafe on Lake Sammamish to display menu items and handle online orders. Used AWS for cloud deployment, Supabase and SQL for data storage, built with Javascript/html
- Worked with the owner of the business as an independent freelancer to determine and meet project requirements.
- Created a secure system with authentication for backend management and integration with Clover
- Increased monthly organic traffic by 4x within 6 months through SEO and improving online presence, gaining over 1000 interactions. Learned to effectively communicate with a client to define project requirements

Undergraduate Researcher, [UW Photonics](#), Seattle, WA | [Github](#)

June. 2026 - Present

- Researched machine learning applications for DSP equalization in fiber optic data communications with microLEDs
- Used Pytorch/Python with a 5000 OFDM waveform dataset to train TCN and LRU channel models
- Developed strong debugging and troubleshooting skills in python
- Designed and simulated a quantized TCN neural network in SystemVerilog to equalize the received signals picked up by a photodiode and achieved BER ~0%/RRMSE 7% with QPSK modulation; meant to run on an FPGA in real time

Web Developer, [Web Impact UW](#), Seattle, WA

Jan. 2026 - June 2026

- Collaborated with a team of 10 developers/designers and a PM to deploy production websites for UW organizations
- Worked within a Git-based workflow including pull requests, code reviews, and collaborative debugging
- Used Nextjs/Vercel for deployment, Sanity for CMS, programmed in javascript/html/css

Electronics and Systems Engineer, [SPARCS](#), Seattle, WA

Jan. 2026 - Present

- Built firmware in C++ (Arduino) to sample analog heart sound data and transmit it wirelessly over Bluetooth BLE
- Developed a Python pipeline to receive BLE packets, apply digital signal filtering, and reconstruct audio output as a .wav file
- Project part of the Dempsey startup competition and funded by The Makers and BWB at UW

Summer Camp Lead, [Open Window School](#), Bellevue, WA

Aug. 2022 - Aug. 2023

- Created and taught a project-based Python curriculum, guiding students through building complete game projects.
- Introduced core CS concepts including control flow, debugging, and algorithmic thinking.
- Communicated with faculty and managed two classes of twelve students each to facilitate an engaging environment

OPEN SOURCE CONTRIBUTIONS

President, [Skyline HS Game Design Club](#), Sammamish, WA

Jan. 2024 - June 2025

- Founded and managed a 50-member student organization, providing learning and shadowing opportunities in game development
- Led weekly club meetings for teaching programming, Unity, and basic game design principles, and maintained a Git organization with multiple collaborators.
- Developed two complete games using Unity and C#

SKILLS & INTERESTS

Strongest Languages: Python, Javascript | **Technical Skills:** Algorithms, NodeJs, NextJs, Css, Bash, Html

Tools: Git, VS Code, IntelliJ, Gemini API, SQL, AWS/Vercel | **Soft Skills:** Communication, Initiative

Interests: Piano, Web Development, Weightlifting

*Referrals available at request